

OBJECT PERMANENCE UNDER SELF-OCCLUSION: BRANCH-STRUCTURED EVIDENCE FOR EMBODIED WORLD-ACTION MODELS

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ABSTRACT

We study robot scene memory and manipulation. The motivating bet is: Make self-occlusion by the robot a distinct world-model failure mode. We introduce a mechanism-level representation that keeps physically admissible but unrealized action outcomes as explicit branches. A synthetic contact-style evaluation shows that the proposed branch mechanism improves hidden-mode success and tail-risk relative to observed-only, augmentation-as-data, and scalar-uncertainty baselines.

1 INTRODUCTION

Robotics papers often collapse unobserved physical alternatives into extra data, variance, or a single predicted next state. For robot scene memory and manipulation, this can erase the difference between modes that are visually similar but action-critical. This paper asks whether a world-action model should expose those alternatives to the planner before execution.

2 HOSTILE PRIOR WORK

The closest hostile records include (pri, 2026a; 2024; 2022; 2025a;b; 2026b). They make generic world models, contact prediction, and uncertainty insufficient as novelty claims. Our boundary is narrower: preserve branch-valued contact futures as the planning object.

3 MECHANISM

Given state s and action a , the model returns an observed next-state prediction plus a finite atlas $B(s, a)$ of admissible latent branches. Planning minimizes nominal loss plus adverse-branch tail risk. This is not bigger data or a new benchmark; it changes the object represented by the WAM.

4 EVIDENCE

We include a runnable synthetic testbed in `src/run_experiment.py`. It compares four variants under nominal and hidden-mode-shift conditions.

Variant	Split	Success	Tail risk
observed_only_wam	nominal	0.487	0.512
observed_only_wam	hidden_mode_shift	0.168	0.833
augmentation_as_data_wam	nominal	0.627	0.373
augmentation_as_data_wam	hidden_mode_shift	0.240	0.760
scalar_uncertainty_wam	nominal	0.657	0.343
scalar_uncertainty_wam	hidden_mode_shift	0.362	0.637
proposed_branch_mechanism	nominal	0.787	0.213
proposed_branch_mechanism	hidden_mode_shift	0.585	0.415

5 LIMITATIONS

The evidence is synthetic. It supports the mechanism boundary and failure mode, not real-robot state of the art. Hardware validation, richer contact geometry, and learned branch discovery remain open.

6 CONCLUSION

Object Permanence Under Self-Occlusion argues that embodied world-action models should preserve unrealized physical alternatives when those alternatives change downstream action value.

REFERENCES

- Self-supervised learning of multi-object keypoints for robotic manipulation, 2022. <http://arxiv.org/abs/2205.08316v2>.
- Vision-language model-driven scene understanding and robotic object manipulation, 2024. <https://doi.org/10.1109/case59546.2024.10711845>.
- Scaling cross-environment failure reasoning data for vision-language robotic manipulation, 2025a. <http://arxiv.org/abs/2512.01946v3>.
- Study of failure mode and ultimate bearing capacity for self-centering sma connection, 2025b. <https://doi.org/10.3390/buildings15183254>.
- Failure-aware rl: Reliable offline-to-online reinforcement learning with self-recovery for real-world manipulation, 2026a. <http://arxiv.org/abs/2601.07821v1>.
- Articulated 3d scene graphs for open-world mobile manipulation, 2026b. <http://arxiv.org/abs/2602.16356v1>.